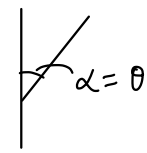
CoursePH410 Course coordinator: Alika Khare

Problem set _06 Due Date: Feb 20, 2025

Note: Please submit the solutions of at least one of the problems (excluding first problem), on thursday, Feb 20, 2025, during the class. The solutions can be submitted in a group of maximum of four students. Name and Roll no. of all the group members should be written clearly on the top. If solutions are found to be reasonably correct then a total of 05 marks will be awarded and in case of non-submission/wrong solution zero marks. List all the boundary conditions as applicable. Where ever applicable draw the well labeled diagram along with axis properly marked.

All the problems in this set are required to be solved via multipole expansion



1. A thin insulating rod carries a line charge density $\lambda = K\cos(\pi z/a)$ for z running from $-a$ to a and K is a constant. Obtain the very first non zero term in the potential at any point **r** with respect to the center of charge distribution.

2. Consider a sphere of radius R (centered at origin) filled with a uniform volume charge density ρ_o on the northern hemisphere and that of $-\rho_o$ on southern hemisphere.
   a. Find the potential in the region $r>R$ by multipole expansion.
   b. From this obtained the very first non zero term.
   c. From part (a) above, find the expression for dipole moment.
   d. Calculate the approximate expression for electric field due to the very first non zero term in the potential.

3. A spherical shell of radius R carries a surface charge density $\sigma(\theta) = K\cos(\theta)$.
   (a) Calculate the dipole moment due to this charge distribution.
   (b) Calculate the potential far away from this sphere and
   (c) Corresponding electric field.
   (d) What is the potential in the limit $r \to \infty$

4. A sphere of radius R carries a charge density $\rho(r,\theta) = K\frac{R}{r^2}(R-2r))\sin\theta$. Calculate the very first non zero term in the potential far from the sphere along the z axis.

5. A circular ring in the xy plane (radius R, centered at the origin) carries a uniform line charge density λ. Find the first three terms (l = 0, 1, 2) in the multipole expansion at any point $V(\mathbf{r},\theta)$.
   ↳ Subtle aspects

6. A charge distribution is specified by $\rho(r,\theta) = \frac{1}{64\pi} r^2 e^{-r} \sin^2\theta$. Using multipole approximation, obtain the expression for potential.

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Tutorial 6

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4. A sphere of radius R carries a charge density $\rho(r,\theta) = k\frac{R}{r^2}(R-2r)sin\theta$ Calculate the very first non zero term in the potential far from the sphere along the z axis.

Solution:
Since $\vec{r}$ is on the z-axis, we have

$$\alpha = \theta'$$

Charge density is given as,

$$\rho(r',\theta') = k\frac{R}{r'^2}(R-2r')\sin\theta'$$

Monopole:

$$V_{\text{mon}}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r}\int\left[k\frac{R}{r'^2}(R-2r')\sin\theta'\right]r'^2\sin\theta' dr' d\theta' d\phi'$$

r' Integral:

$$\int_0^R (R-2r')\,dr' = \left(Rr' - r'^2\right)\Big|_0^R = R^2 - R^2 = 0$$

Thus, the monopole part is

$$V_{mon}(\vec{r}) = 0$$

Dipole:

$$V_{\text{dip}}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\int r'\cos\theta'\left[k\frac{R}{r'^2}(R-2r')\sin\theta'\right]r'^2\sin\theta' dr' d\theta' d\phi'$$

θ' Integral:

$$I = \int_0^\pi \sin^2\theta'\cos\theta' d\theta'$$

Let,

$$u = \sin\theta'$$
$$du = d\theta'\cos\theta'$$

Since

$$\theta' = 0 \Rightarrow u = 0$$
$$\theta' = \pi \Rightarrow u = 0$$

The integral evaluates to:

$$I = \int_0^0 u^2\,du = 0 \quad \text{[Integral becomes zero because upper and lower limits are the same]}$$

Thus, the dipole part is

$$V_{\text{dip}}(\vec{r}) = 0$$

Quadrupole:

$$V_{\text{quad}}(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r^3}\int r'^2\left[k\frac{R}{r'^2}(R-2r')\sin\theta'\right]\left[\frac{3}{2}\cos^2\theta' - \frac{1}{2}\right]r'^2\sin\theta' dr' d\theta' d\phi'$$

$$V_{\text{quad}}(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r^3}\frac{1}{2}kR\int r'^2\left[\frac{1}{r'^2}(R-2r')\sin\theta'\right]\left[3\cos^2\theta' - 1\right]r'^2\sin\theta' dr' d\theta' d\phi'$$

r' Integral:

$$\int_0^R r'^2(R-2r')\,dr' = \left(\frac{r'^3}{3}R - \frac{r'^4}{2}\right)\Big|_0^R = \frac{R^4}{3} - \frac{R^4}{2} = -\frac{R^4}{6}$$

θ' Integral:

$$\int_0^\pi \left(3\cos^2\theta' - 1\right)\sin^2\theta' d\theta' = 2\int_0^\pi \sin^2\theta' d\theta' - 3\int_0^\pi \sin^4\theta' d\theta'$$

$$I_1 = 2\int_0^\pi \sin^2\theta' d\theta' = \int_0^\pi (1 - cos(2\theta'))d\theta' = \int_0^\pi d\theta' - \int_0^\pi cos(2\theta')d\theta' = \pi - \frac{sin(2\theta')}{2}\Big|_0^\pi = \pi - 0 + 0 = \pi$$

$$\begin{aligned} I_2 &= 3\int_0^\pi \sin^4\theta' d\theta' = 3\int_0^\pi \left(\frac{3}{8} - \frac{1}{2}\cos 2\theta' + \frac{1}{8}\cos 4\theta'\right) d\theta' \\ &= 3\left(\frac{3}{8}\int_0^\pi d\theta' - \frac{1}{2}\int_0^\pi \cos 2\theta' d\theta' + \frac{1}{8}\int_0^\pi \cos 4\theta' d\theta'\right) \\ &= 3\left(\frac{3}{8}\pi - \frac{\sin 2\theta'}{4}\Big|_0^\pi + \frac{\sin 4\theta'}{32}\Big|_0^\pi\right) \\ &= 3\left(\frac{3}{8}\pi - 0 + 0\right) = \frac{9}{8}\pi \end{aligned}$$

Thus the θ' integral becomes

$$\int_0^\pi \left(3\cos^2\theta' - 1\right)\sin^2\theta' d\theta' = \pi - \frac{9}{8}\pi = -\frac{\pi}{8}$$

ϕ' integral:

$$\int_0^{2\pi} d\phi = 2\pi$$

$$V_{\text{quad}}(\vec{r}) = \frac{1}{4\pi\varepsilon_0 r^3}\frac{1}{2}kR\left(-\frac{R^4}{6}\right)\left(-\frac{\pi}{8}\right)2\pi = \frac{1}{4\pi\varepsilon_0 r^3}\frac{k\pi^2 R^5}{48}$$

2. Consider a sphere of radius R (centered at origin) filled with a uniform volume charge density ρ_o on the northern hemisphere and that of $-\rho_o$ on southern hemisphere.
   a. Find the potential in the region $r>R$ by multipole expansion.
   b. From this obtained the very first non zero term.
   c. From part (a) above, find the expression for dipole moment.
   d. Calculate the approximate expression for electric field due to the very first non zero term in the potential.

$$\rho(\theta) = \begin{cases} \rho_0 & 0 < \theta < \frac{\pi}{2} \\ -\rho_0 & \frac{\pi}{2} < \theta < \pi \end{cases}$$

$$V_n(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \int (r')^n \rho(r') P_n(\cos\alpha)\, d\tau'$$

$$P_n(\cos\alpha) = \frac{4\pi}{2n+1} \sum_{m=-n}^{n} Y^*_{nm}(\theta', \varphi') Y_{nm}(\theta, \varphi)$$

$$V_n(r) = \frac{1}{4\pi\varepsilon_0} \sum_{n=0}^{\infty} \frac{4\pi}{r^{n+1}} \frac{1}{(2n+1)} \int r^n \rho(r', \theta') \sum_{m=-n}^{n} Y^*_{nm}(\theta', \varphi') Y_{nm}(\theta, \varphi)\, d\tau$$

$$= \frac{1}{4\pi\varepsilon_0} \sum_{n=0}^{\infty} \frac{1}{r^{n+1}} \frac{4\pi}{2n+1} \sum_{m=-n}^{n} Y_{nm}(r, \theta) \int_0^R r'^{n+2}\, dr' \int \rho(\theta') Y^*_{nm}(\theta', \varphi') \sin\theta' d\theta' d\varphi'$$

$$m = 0$$

$$= \frac{1}{4\pi\varepsilon_0} \sum_{m=0}^{\infty} \frac{1}{r^{n+1}} \frac{4\pi}{2n+1} 2\pi Y_{n0}(\theta, \varphi) \int_0^R r'^{n+2}\, dr' \int \rho(\theta') \sin\theta' Y_{n0}(\theta', \varphi')\, d\theta'$$

For $n=0$ (Monopole)

$$V_0(r) = \frac{2\pi}{\varepsilon_0 r} \frac{R^3}{3} \left[\rho_0 \int_0^{\pi/2} \sin\theta' d\theta' - \rho_0 \int_{\pi/2}^{\pi} \sin\theta' d\theta' \right]$$

$$= \frac{2\pi}{\varepsilon_0 r} \frac{R^3}{3} \left[\rho_0 \left. -\cos\theta' \right|_0^{\pi/2} - \rho_0 \left[-\cos\theta' \right]_{\pi/2}^{\pi} \right] = 0$$

For dipole, $n=1$

$$V(r,\theta) = \frac{2\pi}{\varepsilon_0 r^2} Y_{10}(\theta,\varphi) \int_0^R r'^3 dr' \left[\rho_0 \int f(\theta) \cos\theta' \sin\theta' d\theta' \right]$$

$$= \frac{2\pi}{\varepsilon_0 r^2} \cos\theta \sqrt{\frac{3}{4\pi}} \sqrt{\frac{3}{4\pi}} \frac{R^4}{4} \left[\rho_0 \int_0^{\pi/2} \cos\theta' \sin\theta' d\theta' \right]$$

$$= \frac{\pi \rho_0 R^4}{8 \pi \varepsilon_0 r^2} \cos\theta$$

$$E = -\vec{\nabla} V_1(\vec{r})$$

$$\vec{p} = \frac{\pi \rho_0 R^4}{2 r^3}$$

$$\frac{p}{4\pi\varepsilon_0 r^2} (2\cos\theta \hat{r} + \sin\theta \hat{\theta})$$